STRATA shear data
Link to physics

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1 Equations solved and notation

The total velocity $\boldsymbol{u} = \bar{u}(z)\hat{\boldsymbol{x}} + \boldsymbol{u}'(\boldsymbol{x}, t)$ comprises of a mean vertical shear $\bar{u}(z) = zS$ (shear rate $= S$) and three-dimensional unsteady perturbations $\boldsymbol{u}'$, noting that $\boldsymbol{x} = (x, y, z)$.

The density field $\rho = \rho_0 + \bar{\rho}(z) + \rho'(\boldsymbol{x}, t)$ has a reference density ρ_0, a linearly stratified background $\bar{\rho}(z) = -(\rho_0/g)N^2 z$ with buoyancy frequency N, and perturbations ρ'.

The equations solved, following the notation used by the solver, are:

$$\nabla \cdot \boldsymbol{u}' = 0, \tag{1a}$$

$$\frac{\partial \boldsymbol{u}'}{\partial t} + (\boldsymbol{u}' \cdot \nabla)\boldsymbol{u}' = -\frac{1}{\rho_0}\nabla p' - z\frac{d\bar{u}}{dz}\frac{\partial \boldsymbol{u}'}{\partial x} - w\frac{d\bar{u}}{dz}\hat{\boldsymbol{x}} - \frac{g}{\rho_0}\rho'\,\hat{\boldsymbol{z}} + \nu\,\nabla \cdot \nabla \boldsymbol{u}', \tag{1b}$$

$$\frac{\partial \rho'}{\partial t} + (\boldsymbol{u}' \cdot \nabla)\rho' = -z\frac{d\bar{u}}{dz}\frac{\partial \rho'}{\partial x} - \frac{d\bar{\rho}}{dz}w' + D\nabla \cdot \nabla \rho', \tag{1c}$$

The solver needs to know: $\nu, D, \rho_0, g, \dfrac{d\bar{u}}{dz}, \dfrac{d\bar{\rho}}{dz}$ as well as the box size L_x, L_y, L_z and grid size N_x, N_y, N_z, which are given below.

2 Metadata and physical conversion

Here we explain how the `variable names` provided in the netCDF parameter files and the Excel spreadsheet compare to the variables used in the paper, which also helps clarify the link between the numerical methodology and the flow physics.

`name`, simply R1P1, R1P7, etc, see main table.

`Lx` $= L_x = 8\pi$, `Ly`$= L_y = 4\pi$, `Lz` $= L_z = 2\pi$, fixed in all runs.

`tStamp` time stamp (snapshot) at which the simulation was converged and saved.

`dRef` $= \rho_0 = 1$ fixed in all runs.

`dUdz` $= \dfrac{d\bar{u}}{dz} = S = 1$ fixed in all runs.

`dGrad` $= \dfrac{d\rho}{dz} = -0.001$ fixed in all runs.

`zAccel` $= g$, varied slightly during the simulation following the method described in the paper until convergence to a stationary state. The values provided is the converged stationary value. Since $N^2 = -\dfrac{g}{\rho_0}\dfrac{d\bar{\rho}}{dz} = 0.001g$ and $\mathrm{Ri} = N^2/S^2 = N^2$ we find that $g = 1000\,\mathrm{Ri}$, hence $g/1000$ simply gives the converged Ri which is reported in the main table.

`targKE` $= E_k = \langle (u')^2 + (v')^2 + (w')^2 \rangle /2$, which is the target value and (to an excellent approximation) the converged stationary value.

`kinV` $= \nu$ varies between runs, this is decided by trial and error to achieve a certain Re_b. The actual Re_b value reported in the main table (close to the one targetted) is simply the measured stationary $\langle \varepsilon \rangle$ divided by this fixed `kinV` $= \nu$ and by the stationary $N^2 = 0.001g$ ($= 0.001\times$ `zAccel`). This `kinV` $= \nu$ is what ultimately controls the dynamic range and the value of the Kolmogorov scale. The stationary Froude number $\mathrm{Fr} = \dfrac{\langle \varepsilon \rangle}{N E_K}$ reported in the table is then calculated from this stationary $\langle \varepsilon \rangle$ divided by the stationary $N = \sqrt{0.001g}$ and the target E_k.

`Pr` $= \mathrm{Pr}$, varies between runs ($= 1, 7$ or 50), giving the solver $D = \nu/\mathrm{Pr}$. This will control the separation between the Kolmogorov and Batchelor scale.

`Nx` $= N_x$, `Ny` $= N_y$, `Nz` $= N_z$, vary between runs but always $N_x = 2N_y = 4N_z$). The N_x value (in other words, the grid size $\Delta = L_x/N_x$) is chosen to guarantee that the product of the maximum wavenumber resolved and Batchelor lengthscale satisfy $\kappa_{max} L_B \geq 2.0$ (in other words $\Delta \leq (\pi/2) L_B$). This ensures that we have true DNS. An approximate value of L_B can be calculated a priori from the target Re_b, the expected Fr and the set Pr, since we enforce $L_x = 40 L_L$ and by definition $L_L = \mathrm{Fr}^{-3/2} L_O = \mathrm{Fr}^{-3/2}\mathrm{Re}_\mathrm{b}^{3/4} L_K = \mathrm{Fr}^{-3/2}\mathrm{Re}_\mathrm{b}^{3/4}\mathrm{Pr}^{1/2}$. Thus the minimum allowable value is:

$$N_x = \frac{L_x}{\Delta} = \frac{80}{\pi}\,\mathrm{Fr}^{-3/2}\,\mathrm{Re}_\mathrm{b}^{3/4}\,\mathrm{Pr}^{1/2} \approx 80\,\mathrm{Re}_\mathrm{b}^{3/4}\,\mathrm{Pr}^{1/2}, \tag{1}$$

since the shear-forced dynamics adjust to give a typical $\mathrm{Fr}^{-3/2} \approx \pi$. This minimum N_x rule of thumb (at fixed Fr) is plotted in the main paper with gray contours on figure 2a. The actual values N_x reported in the main table are based on the actual measured Fr.

3 Conversion of density to buoyancy

As we make the Boussinesq approximation, we assumed that ρ' only plays a dynamical role when multiplied by g on the right hand side of the momentum equation. Thus, dynamically, it makes sense to consider not ρ' per se, but the acceleration caused by those perturbations around the neutral density ρ_0. This leads us to defining the buoyancy field (positive when fluid is lighter than ρ_0 and vice versa) as:

$$b = \frac{-g(\rho - \rho_0)}{\rho_0} = N^2 z - \frac{g}{\rho_0}\rho' = \bar{b}(z) + b'(\boldsymbol{x}, t) \tag{2}$$

We see that it comprises the mean stratification $\bar{b} = N^2 z$ (vertical buoyancy gradient $d\bar{b}/dz = N^2$) and buoyancy turbulent fluctuations $b' = (-g/\rho_0)\rho'$. The governing equations in terms of buoyancy are more elegant, given in the main text, and repeated below for convenience:

$$\nabla \cdot \boldsymbol{u}' = 0, \tag{1a}$$

$$\frac{\partial \boldsymbol{u}'}{\partial t} + (\boldsymbol{u}' \cdot \nabla)\boldsymbol{u}' = -\frac{1}{\rho_0}\nabla p' - zS\frac{\partial \boldsymbol{u}'}{\partial x} - wS\hat{\boldsymbol{x}} + b'\,\hat{\boldsymbol{z}} + \nu\nabla^2\boldsymbol{u}', \tag{1b}$$

$$\frac{\partial b'}{\partial t} + (\boldsymbol{u}' \cdot \nabla)b' = -zS\frac{\partial b'}{\partial x} + N^2 w' + D\nabla^2 b', \tag{1c}$$

4 Main three-dimensional fields

Here we list the six `variables provided on the repository` at the final time stamp (stationary state) and the variables we plotted in the figures (with sensible normalization).
`u` $= u'(x,y,z)$, `v` $= v'(x,y,z)$, `w` $= w'(x,y,z)$, `r` $= \rho'(x,y,z)$

The magnitude of velocity fluctuations is $E_k^{1/2}$, so we plotted $(u',v',w')/E_k^{1/2}$, i.e. normalized by root mean square and immediately relatable to the physical E_k quoted).

Total velocity field is $u = Sz + u' = z + u'$ where $z \in [-L_z/2, L_z/2] = [-\pi/2, \pi/2]$.

The buoyancy fluctuation field is computed as $b' = (-g/\rho_0)\rho' = -(\texttt{zAccel}/\texttt{dRef}) * \texttt{r} = -\texttt{zAccel} * \texttt{r}$.

Total buoyancy field is $b = N^2 z + b'$, remembering that $N^2 = 0.001g$ here.

The energy of buoyancy fluctuations is $= \langle (b')^2 \rangle /2 = N^2 E_p$ by definition of the (available) potential energy E_p. It is sensible to plot $b'/(NE_p^{1/2})$ to normalize it by its root mean square which is immediately relatable to the physical N and $E_p = E_k R_{PK}$.

Dissipation rate of turbulent kinetic energy: `ee` $= \varepsilon(x,y,z) = 2\nu\, s'_{ij} s'_{ij}$ where the strain rate tensor is $s'_{ij} = \frac{1}{2}(\partial_j u'_i + \partial_i u'_j)$. Written in full we get:

$$\begin{aligned}\varepsilon = 2\nu\Big[&(\partial_x u')^2 + (\partial_y v')^2 + (\partial_z w')^2 \\ &+ \frac{1}{2}(\partial_y u' + \partial_x v')^2 + \frac{1}{2}(\partial_z u' + \partial_x w')^2 + \frac{1}{2}(\partial_z v' + \partial_y w')^2\Big]\end{aligned} \tag{3}$$

$$\begin{aligned}= \nu\Big[&2(\partial_x u')^2 + 2(\partial_y v')^2 + 2(\partial_z w')^2 \\ &+ (\partial_y u')^2 + (\partial_x v')^2 + 2\,\partial_y u'\,\partial_x v' \\ &+ (\partial_z u')^2 + (\partial_x w')^2 + 2\,\partial_z u'\,\partial_x w' \\ &+ (\partial_z v')^2 + (\partial_y w')^2 + 2\,\partial_z v'\,\partial_y w'\Big].\end{aligned} \tag{4}$$

Here is makes sense to plot $\log_{10}(\varepsilon/\langle\varepsilon\rangle)$, i.e. the true dissipation normalized by its volume average, which varies between runs is easily connected to the physics through $\mathrm{Re_b}$, ν and N^2.

Mixing rate (dissipation of turbulent potential energy): `chi` $= \chi(x,y,z) = \dfrac{D}{N^2}|\nabla b'|^2 = \dfrac{D}{N^2}\Big[(\partial_x b')^2 + (\partial_y b')^2 + (\partial_z b')^2\Big]$, remembering that $N^2 = 0.001g$ here. Here is makes sense to plot $\log_{10}(\chi/\langle\chi\rangle)$, as $\langle\chi\rangle$ varies between runs but is easily connected to the physics through ε and $\Gamma = \dfrac{\langle\chi\rangle}{\langle\varepsilon\rangle}$ (provided in the table).